

Filtering and Ordering Columns & Rows

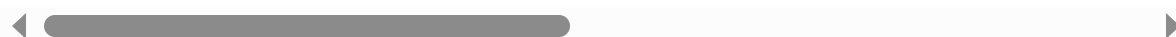
In [17]: `import pandas as pd`

In [18]: `df = pd.read_csv(r"world_population.csv")`
`df`

Out[18]:

	Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	2015 Population	Pe
0	36	AFG	Afghanistan	Kabul	Asia	41128771.0	38972230.0	33753499.0	28
1	138	ALB	Albania	Tirana	Europe	2842321.0	2866849.0	2882481.0	2
2	34	DZA	Algeria	Algiers	Africa	44903225.0	43451666.0	39543154.0	35
3	213	ASM	American Samoa	Pago Pago	Oceania	44273.0	46189.0	51368.0	
4	203	AND	Andorra	Andorra la Vella	Europe	79824.0	77700.0	71746.0	
...	...	...	...	...	...	...	...	...	...
229	226	WLF	Wallis and Futuna	Mata-Utu	Oceania	11572.0	11655.0	12182.0	
230	172	ESH	Western Sahara	El Aaiún	Africa	575986.0	556048.0	491824.0	
231	46	YEM	Yemen	Sanaa	Asia	33696614.0	32284046.0	28516545.0	24
232	63	ZMB	Zambia	Lusaka	Africa	20017675.0	18927715.0	NaN	13
233	74	ZWE	Zimbabwe	Harare	Africa	16320537.0	15669666.0	14154937.0	12

234 rows × 17 columns



In [31]: `df.columns`

Out[31]: Index(['Rank', 'CCA3', 'Country', 'Capital', 'Continent', '2022 Population', '2020 Population', '2015 Population', '2010 Population', '2000 Population', '1990 Population', '1980 Population', '1970 Population', 'Area (km²)', 'Density (per km²)', 'Growth Rate', 'World Population Percentage'], dtype='str')

In [32]: `df[df['Rank'] <= 10]`

Out[32]:

	Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	Popi
16	8	BGD	Bangladesh	Dhaka	Asia	1.711864e+08	1.674210e+08	1.57830
27	7	BRA	Brazil	Brasilia	South America	2.153135e+08	2.131963e+08	2.05188
41	1	CHN	China	Beijing	Asia	1.425887e+09	1.424930e+09	1.39377
92	2	IND	India	New Delhi	Asia	1.417173e+09	1.396387e+09	1.32288
93	4	IDN	Indonesia	Jakarta	Asia	2.755013e+08	2.718580e+08	2.59092
131	10	MEX	Mexico	Mexico City	North America	1.275041e+08	1.259983e+08	1.20149
149	6	NGA	Nigeria	Abuja	Africa	2.185412e+08	2.083274e+08	1.83995
156	5	PAK	Pakistan	Islamabad	Asia	2.358249e+08	2.271967e+08	2.10965
171	9	RUS	Russia	Moscow	Europe	1.447133e+08	1.456173e+08	1.44668
221	3	USA	United States	Washington, D.C.	North America	3.382899e+08	3.359420e+08	3.24607

In [33]: df[df['Rank'] <= 10].shape

Out[33]: (10, 17)

In [54]: df[df['Country'].isin(['Bangladesh', 'Brazil'])]

Out[54]:

	Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	2015 Population	I
16	8	BGD	Bangladesh	Dhaka	Asia	171186372.0	167420951.0	157830000.0	1.
27	7	BRA	Brazil	Brasilia	South America	215313498.0	213196304.0	205188205.0	1.

In [35]: df[df['Country'].str.contains('United')]

Out[35]:

	Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	2015 Population
219	97	ARE	United Arab Emirates	Abu Dhabi	Asia	9441129.0	9287289.0	8916899.0
220	21	GBR	United Kingdom	London	Europe	67508936.0	67059474.0	65224364.0
221	3	USA	United States	Washington, D.C.	North America	338289857.0	335942003.0	324607776.0
222	200	VIR	United States Virgin Islands	Charlotte Amalie	North America	99465.0	100442.0	102803.0



```
In [36]: df2 = df.set_index('Country') #index applied "Country"
df2
```

Out[36]:

	Rank	CCA3	Capital	Continent	2022 Population	2020 Population	2015 Population	2010 Population
Country								
Afghanistan	36	AFG	Kabul	Asia	41128771.0	38972230.0	33753499.0	281896
Albania	138	ALB	Tirana	Europe	2842321.0	2866849.0	2882481.0	29133
Algeria	34	DZA	Algiers	Africa	44903225.0	43451666.0	39543154.0	358563
American Samoa	213	ASM	Pago Pago	Oceania	44273.0	46189.0	51368.0	548
Andorra	203	AND	Andorra la Vella	Europe	79824.0	77700.0	71746.0	715
...	...	...	...	...	...	...	...	...
Wallis and Futuna	226	WLF	Mata-Utu	Oceania	11572.0	11655.0	12182.0	131
Western Sahara	172	ESH	El Aaiún	Africa	575986.0	556048.0	491824.0	4132
Yemen	46	YEM	Sanaa	Asia	33696614.0	32284046.0	28516545.0	247439
Zambia	63	ZMB	Lusaka	Africa	20017675.0	18927715.0	NaN	137920
Zimbabwe	74	ZWE	Harare	Africa	16320537.0	15669666.0	14154937.0	128397

234 rows × 16 columns



```
In [37]: df2.filter(items = ['Continent','CCA3'], axis = 1)
```

Out[37]:

		Continent	CCA3
Country			
Afghanistan	Asia	AFG	
Albania	Europe	ALB	
Algeria	Africa	DZA	
American Samoa	Oceania	ASM	
Andorra	Europe	AND	
...	...	...	
Wallis and Futuna	Oceania	WLF	
Western Sahara	Africa	ESH	
Yemen	Asia	YEM	
Zambia	Africa	ZMB	
Zimbabwe	Africa	ZWE	

234 rows × 2 columns

```
In [38]: df2.filter(items = ['Zimbabwe'], axis = 0)
```

Out[38]:

	Rank	CCA3	Capital	Continent	2022 Population	2020 Population	2015 Population	2010 Population
Zimbabwe	74	ZWE	Harare	Africa	16320537.0	15669666.0	14154937.0	12839771

```
In [39]: df2.filter(like = 'United', axis = 0)
```

Out[39]:

	Rank	CCA3	Capital	Continent	2022 Population	2020 Population	2015 Population	Pc
Country								
United Arab Emirates	97	ARE	Abu Dhabi	Asia	9441129.0	9287289.0	8916899.0	8
United Kingdom	21	GBR	London	Europe	67508936.0	67059474.0	65224364.0	62
United States	3	USA	Washington, D.C.	North America	338289857.0	335942003.0	324607776.0	311
United States Virgin Islands	200	VIR	Charlotte Amalie	North America	99465.0	100442.0	102803.0	

```
In [55]: df2.loc[df2['Continent'] == 'Asia']
```

Out[55]:

	Rank	CCA3	Capital	Continent	2022 Population	2020 Population	2019 Populat
Country							
Afghanistan	36	AFG	Kabul	Asia	4.112877e+07	3.897223e+07	3.375350e-
Armenia	140	ARM	Yerevan	Asia	2.780469e+06	2.805608e+06	2.878595e-
Azerbaijan	91	AZE	Baku	Asia	1.035807e+07	NaN	9.863480e-
Bahrain	154	BHR	Manama	Asia	1.472233e+06	1.477469e+06	1.362142e-
Bangladesh	8	BGD	Dhaka	Asia	1.711864e+08	1.674210e+08	1.578300e-
Bhutan	165	BTN	Thimphu	Asia	7.824550e+05	7.725060e+05	7.432740e-
Brunei	175	BRN	Bandar Seri Begawan	Asia	4.490020e+05	4.417250e+05	4.214370e-
Cambodia	73	KHM	Phnom Penh	Asia	1.676784e+07	1.639686e+07	1.541752e-
China	1	CHN	Beijing	Asia	1.425887e+09	1.424930e+09	1.393715e-
Georgia	131	GEO	Tbilisi	Asia	3.744385e+06	3.765912e+06	3.771132e-
Hong Kong	104	HKG	Hong Kong	Asia	7.488865e+06	7.500958e+06	7.399838e-
India	2	IND	New Delhi	Asia	1.417173e+09	1.396387e+09	1.322867e-
Indonesia	4	IDN	Jakarta	Asia	2.755013e+08	2.718580e+08	2.590920e-
Iran	17	IRN	Tehran	Asia	8.855057e+07	8.729019e+07	8.179084e-
Iraq	35	IRQ	Baghdad	Asia	4.449612e+07	4.255698e+07	3.775781e-
Israel	98	ISR	Jerusalem	Asia	9.038309e+06	8.757489e+06	8.007778e-
Japan	11	JPN	Tokyo	Asia	1.239517e+08	1.252448e+08	1.272509e-
Jordan	83	JOR	Amman	Asia	1.128587e+07	1.092872e+07	9.494246e-
Kazakhstan	66	KAZ	Nursultan	Asia	1.939800e+07	1.897924e+07	1.783591e-
Kuwait	129	KWT	Kuwait City	Asia	4.268873e+06	4.360444e+06	3.908743e-
Kyrgyzstan	110	KGZ	Bishkek	Asia	6.630623e+06	6.424874e+06	5.914980e-
Laos	103	LAO	Vientiane	Asia	7.529475e+06	7.319399e+06	6.787419e-
Lebanon	119	LBN	Beirut	Asia	5.489739e+06	5.662923e+06	6.398940e-
Macau	167	MAC	Concelho de Macau	Asia	6.951680e+05	6.762830e+05	6.152390e-
Malaysia	45	MYS	Kuala Lumpur	Asia	3.393822e+07	3.319999e+07	3.106883e-
Maldives	174	MDV	Malé	Asia	5.237870e+05	5.144380e+05	4.355820e-

	Rank	CCA3	Capital	Continent	2022 Population	2020 Population	21 Populat
Country							
Mongolia	134	MNG	Ulaanbaatar	Asia	3.398366e+06	3.294335e+06	2.964749e-
Myanmar	26	MMR	Nay Pyi Taw	Asia	5.417931e+07	5.342320e+07	5.148395e-
Nepal	49	NPL	Kathmandu	Asia	3.054758e+07	2.934863e+07	2.761032e-
North Korea	56	PRK	Pyongyang	Asia	2.606942e+07	2.586747e+07	2.525802e-
Oman	127	OMN	Muscat	Asia	4.576298e+06	4.543399e+06	4.191776e-
Pakistan	5	PAK	Islamabad	Asia	2.358249e+08	2.271967e+08	2.109693e-
Palestine	122	PSE	Ramallah	Asia	5.250072e+06	5.019401e+06	4.484614e-
Philippines	13	PHL	Manila	Asia	1.155590e+08	1.121910e+08	1.030314e-
Qatar	143	QAT	Doha	Asia	2.695122e+06	2.760385e+06	2.414573e-
Saudi Arabia	41	SAU	Riyadh	Asia	3.640882e+07	3.599711e+07	3.274985e-
Singapore	113	SGP	Singapore	Asia	5.975689e+06	5.909869e+06	5.650018e-
South Korea	29	KOR	Seoul	Asia	5.181581e+07	5.184469e+07	5.099440e-
Sri Lanka	61	LKA	Colombo	Asia	2.183214e+07	2.171508e+07	2.133670e-
Syria	60	SYR	Damascus	Asia	2.212525e+07	2.077260e+07	1.920518e-
Taiwan	57	TWN	Taipei	Asia	2.389339e+07	2.382146e+07	2.351214e-
Tajikistan	95	TJK	Dushanbe	Asia	9.952787e+06	9.543207e+06	8.524063e-
Thailand	20	THA	Bangkok	Asia	7.169703e+07	7.147566e+07	7.029440e-
Timor-Leste	155	TLS	Dili	Asia	NaN	1.299995e+06	1.205813e-
Turkey	18	TUR	Ankara	Asia	8.534124e+07	8.413543e+07	7.964618e-
Turkmenistan	111	TKM	Ashgabat	Asia	6.430770e+06	6.250438e+06	5.766431e-
United Arab Emirates	97	ARE	Abu Dhabi	Asia	9.441129e+06	9.287289e+06	8.916899e-
Uzbekistan	43	UZB	Tashkent	Asia	3.462765e+07	3.352666e+07	3.094942e-
Vietnam	16	VNM	Hanoi	Asia	9.818686e+07	9.664868e+07	9.219140e-
Yemen	46	YEM	Sanaa	Asia	3.369661e+07	3.228405e+07	2.851654e-

In [41]: df

Out[41]:

	Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	2015 Population	Pc
0	36	AFG	Afghanistan	Kabul	Asia	41128771.0	38972230.0	33753499.0	28
1	138	ALB	Albania	Tirana	Europe	2842321.0	2866849.0	2882481.0	2
2	34	DZA	Algeria	Algiers	Africa	44903225.0	43451666.0	39543154.0	35
3	213	ASM	American Samoa	Pago Pago	Oceania	44273.0	46189.0	51368.0	
4	203	AND	Andorra	Andorra la Vella	Europe	79824.0	77700.0	71746.0	
...	...	...	...	...	...	...	...	...	...
229	226	WLF	Wallis and Futuna	Mata-Utu	Oceania	11572.0	11655.0	12182.0	
230	172	ESH	Western Sahara	El Aaiún	Africa	575986.0	556048.0	491824.0	
231	46	YEM	Yemen	Sanaa	Asia	33696614.0	32284046.0	28516545.0	24
232	63	ZMB	Zambia	Lusaka	Africa	20017675.0	18927715.0	NaN	13
233	74	ZWE	Zimbabwe	Harare	Africa	16320537.0	15669666.0	14154937.0	12

234 rows × 17 columns



In [42]: df2.filter(like = 'United', axis = 0)

Out[42]:

	Rank	CCA3	Capital	Continent	2022 Population	2020 Population	2015 Population	Pc
Country								
United Arab Emirates	97	ARE	Abu Dhabi	Asia	9441129.0	9287289.0	8916899.0	8
United Kingdom	21	GBR	London	Europe	67508936.0	67059474.0	65224364.0	62
United States	3	USA	Washington, D.C.	North America	338289857.0	335942003.0	324607776.0	311
United States Virgin Islands	200	VIR	Charlotte Amalie	North America	99465.0	100442.0	102803.0	



```
In [43]: df2.loc['United States']
```

```
Out[43]: Rank                3
CCA3                USA
Capital    Washington, D.C.
Continent    North America
2022 Population    338289857.0
2020 Population    335942003.0
2015 Population    324607776.0
2010 Population    311182845.0
2000 Population    282398554.0
1990 Population    248083732.0
1980 Population    223140018.0
1970 Population    200328340.0
Area (km²)        9372610.0
Density (per km²)    36.0935
Growth Rate        1.0038
World Population Percentage    4.24
Name: United States, dtype: object
```

```
In [46]: #df2 = df2.reset_index()
#df2[df2['Country'] == 'United States'] # when we havenot made any index
```

```
In [45]: df2.iloc[3]
```

```
Out[45]: Rank                213
CCA3                ASM
Capital    Pago Pago
Continent    Oceania
2022 Population    44273.0
2020 Population    46189.0
2015 Population    51368.0
2010 Population    54849.0
2000 Population    58230.0
1990 Population    47818.0
1980 Population    32886.0
1970 Population    27075.0
Area (km²)        199.0
Density (per km²)    222.4774
Growth Rate        0.9831
World Population Percentage    0.0
Name: American Samoa, dtype: object
```

```
In [52]: df.sort_values(by=['Continent', 'Country', 'CCA3'], ascending=[False, True, False]) #
```

Out[52]:

	Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	2015 Population
8	33	ARG	Argentina	Buenos Aires	South America	45510318.0	45036032.0	43257065.0
24	80	BOL	Bolivia	Sucre	South America	12224110.0	11936162.0	11090085.0
27	7	BRA	Brazil	Brasilia	South America	215313498.0	213196304.0	205188205.0
40	65	CHL	Chile	Santiago	South America	19603733.0	19300315.0	17870124.0
42	28	COL	Colombia	Bogota	South America	51874024.0	50930662.0	47119728.0
...	...	...	...	...	...	...	...	...
212	79	TUN	Tunisia	Tunis	Africa	12356117.0	12161723.0	11557779.0
217	31	UGA	Uganda	Kampala	Africa	47249585.0	44404611.0	37477356.0
230	172	ESH	Western Sahara	El Aaiún	Africa	575986.0	556048.0	491824.0
232	63	ZMB	Zambia	Lusaka	Africa	20017675.0	18927715.0	NaN
233	74	ZWE	Zimbabwe	Harare	Africa	16320537.0	15669666.0	14154937.0

234 rows × 17 columns



In [53]: `df[df['Rank'] < 10].sort_values(by=['Rank'],ascending=[True]) #sorting on Rank colu`

Out[53]:

	Rank	CCA3	Country	Capital	Continent	2022 Population	2020 Population	Popu
41	1	CHN	China	Beijing	Asia	1.425887e+09	1.424930e+09	1.3937
92	2	IND	India	New Delhi	Asia	1.417173e+09	1.396387e+09	1.3228
221	3	USA	United States	Washington, D.C.	North America	3.382899e+08	3.359420e+08	3.2460
93	4	IDN	Indonesia	Jakarta	Asia	2.755013e+08	2.718580e+08	2.5909
156	5	PAK	Pakistan	Islamabad	Asia	2.358249e+08	2.271967e+08	2.1096
149	6	NGA	Nigeria	Abuja	Africa	2.185412e+08	2.083274e+08	1.8399
27	7	BRA	Brazil	Brasilia	South America	2.153135e+08	2.131963e+08	2.0518
16	8	BGD	Bangladesh	Dhaka	Asia	1.711864e+08	1.674210e+08	1.5783
171	9	RUS	Russia	Moscow	Europe	1.447133e+08	1.456173e+08	1.4466

In []: